

Expert Review: Cosmological Perturbation Theory from Matter to Galaxies

Lecture notes for three 90-minute lectures and associated exercise solutions

Technical review from the perspective of perturbation theory, EFTofLSS, and galaxy bias

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Executive assessment

The notes are unusually strong for a 20-page draft. They have a coherent conceptual arc - relativistic perturbations to Newtonian dynamics, Eulerian and Lagrangian perturbation theory, EFT completion, galaxy bias, redshift-space distortions, IR resummation, and likelihood-level inference. The notation is mostly consistent, the core one-loop formulas are correct, the diagrams and figures are useful, and the prose is substantially better than the average set of specialist lecture notes.

They should nevertheless **not be taught or distributed in their current form without revision**. The main reasons are:

- 1. The advertised scope is not executable in three 90-minute lectures for the stated audience.**
The plans allocate all 90 minutes to uninterrupted content. There is no time for questions, derivations on the board, conceptual checks, or recovery from a difficult point. The document is a good compact reference or pre-read, but not yet a realistic lecture script.
- 2. Several statements are technically misleading enough to change a student's understanding.**
The most important examples are the notation around Eqs. (1.18)-(1.19), the description of the non-EdS correction to the full F_2 kernel, the claim that SPT has no expansion parameter, the post-shell-crossing interpretation of the Zel'dovich figure, the definition of f_2 when $D_2 < 0$, the treatment of EFT time locality, the missing coefficient of the higher-derivative bias operator, and the statement that full shape by itself measures f rather than $f\sigma_8$.

3. **The claimed end-to-end connection to survey measurements stops one major operator too early.**
The theory multipoles must still be bin-averaged and passed through the survey measurement operator: window convolution or the corresponding deconvolved-estimator response, integral-constraint corrections, observational-systematic templates, and often redshift/light-cone averaging. Without this, the final recipe is not actually the model compared to data.
4. **The exercises do not test the central content of Lecture 2 or most of Lecture 3.** There is no exercise on EFT matching or renormalization, bias renormalization, IR resummation, the Alcock-Paczynski mapping, or the survey window. Exercise 3.2 is not fully solved in the solution set.

The document can become an excellent advanced mini-course after one substantial revision. The necessary changes are finite and concrete; they do not require changing its overall architecture.

Severity convention

- **Must fix:** mathematically wrong, internally inconsistent, or likely to produce a materially wrong physical picture.
- **Major improvement:** not strictly false in a restricted convention, but misleading, incomplete, or incompatible with the stated learning goal.
- **Moderate improvement:** important precision, scope, or pedagogy issue.
- **Minor/editorial:** clarity, notation, layout, or reference improvement.

Highest-priority corrections

1. Clarify the perturbative fields and normalize the velocity divergence

Location: p. 5, Eqs. (1.18)-(1.19); Appendix A; p. 14, Eqs. (3.7)-(3.9).

Severity: Must fix.

Equation (1.18) defines time-independent coefficients δ_n and θ_n through powers of a , whereas Eq. (1.19) immediately writes $\delta^{(n)}$ as a convolution of full-time linear fields $\delta^{(1)}$. The density formula can be inferred, but the corresponding statement for $\theta^{(n)}$ is dimensionally incomplete: G_n is the kernel of a *normalized* velocity divergence, not of θ without an overall factor of $-\mathcal{H}f$.

Use a single convention throughout. The cleanest is

$$\Theta \equiv -\frac{\theta}{\mathcal{H}f}, \quad \delta(\mathbf{k}, \tau) = \sum_{n \geq 1} D_+^n(\tau) \delta_n(\mathbf{k}), \quad \Theta(\mathbf{k}, \tau) = \sum_{n \geq 1} D_+^n(\tau) \Theta_n(\mathbf{k}),$$

$$\delta_n(\mathbf{k}) = \int_{\mathbf{q}_1 \dots \mathbf{q}_n} (2\pi)^3 \delta_D(\mathbf{k} - \mathbf{q}_1 \dots \mathbf{q}_n) F_n(\mathbf{q}_1, \dots, \mathbf{q}_n) \prod_i \delta_0(\mathbf{q}_i),$$

and analogously for Θ_n with G_n . In EdS, $D_+ = a$ and $f = 1$. This convention also makes $\nabla^2 \Phi_v = \Theta$ in Lecture 3 transparent.

2. Correct the statement about the non-EdS second-order kernel

Location: p. 5, paragraph after the three kernel limits.

Severity: Must fix.

The quoted factor

$$1 + \frac{3}{17} (\Omega_m^{-2/63} - 1)$$

is not a uniform multiplicative correction to the full $F_2(\mathbf{q}_1, \mathbf{q}_2)$ kernel. It is associated with a particular second-order growth coefficient or angle-averaged vertex. The exact non-EdS kernel has weakly time-dependent coefficients multiplying distinct angular structures.

Replace the sentence with: the exact kernels are mildly time dependent outside EdS; the quoted fit applies to the angle-averaged second-order vertex, while the lecture adopts the usual EdS-kernel approximation F_n, G_n with the exact linear growth. State the expected accuracy as an approximation, not an identity.

3. Preserve exact IR safety while explaining BAO smearing

Location: pp. 6-7 and p. 15; Exercise 1.3 and its solution.

Severity: Must fix conceptually.

A strict $q \rightarrow 0$ adiabatic displacement cancels in any equal-time correlator, whether or not the spectrum contains BAO. A sharp feature does not violate the equivalence principle. The BAO effect comes from modes that are soft relative to the hard wavenumber k but not soft relative to the feature scale k_{osc} ; the Taylor expansion in q/k_{osc} is not uniformly small, and an infinite set of finite IR-enhanced terms must be resummed.

Replace phrases such as “the cancellation is exact for smooth broadband power but not for sharp features” and “it does not cancel for sharp features” by:

The strict uniform-displacement contribution cancels. For a feature with scale k_{osc} , modes with $q \gtrsim k_{\text{osc}}$ can still be soft relative to k and generate enhanced finite terms. Their cumulative effect is BAO damping and is resummed.

4. Do not use the Zel’dovich approximation beyond first shell crossing as physical dynamics

Location: p. 8, Fig. 2 and surrounding text.

Severity: Must fix.

The sequence cube $\rightarrow$ sheet $\rightarrow$ filament $\rightarrow$ halo is visually effective but physically overclaims what ZA predicts. ZA ceases to be a valid single-stream solution at the first caustic. Later zeros of $1 - D_+ \lambda_2$ and $1 - D_+ \lambda_3$ are formal extrapolations of the deformation map, not predictions of post-shell-crossing evolution, binding, or virialization. A Zel’dovich “knot” is not a halo.

Revise the caption and text to say that ZA robustly predicts the first-collapse morphology and the skeleton of the cosmic web; subsequent stages illustrate the eigenvalue ordering only. Add that adhesion, phase-space methods, or N -body dynamics are needed after the first shell crossing.

Also qualify “only the first collapse is guaranteed.” It is guaranteed only for a selected patch with a sufficiently positive largest eigenvalue and sufficient subsequent growth; it is not guaranteed at an arbitrary point in a late-time accelerating cosmology.

5. Define the second-order growth rate using $|D_2|$

Location: p. 9, Eq. (2.8).

Severity: Must fix.

The notes define $f_i = d \ln D_i / d \ln a$ while using the convention $D_2 < 0$. A real logarithm of D_2 is undefined. Write

$$f_2 \equiv \frac{d \ln |D_2|}{d \ln a} = \frac{\dot{D}_2}{\mathcal{H} D_2}.$$

The velocity formula then remains unchanged.

6. Rewrite “SPT has no expansion parameter”

Location: pp. 9-10, Sec. 2.3.

Severity: Must fix.

This section contradicts the earlier and later statements that loops are ordered by Δ^2 . At $k \ll k_{\text{NL}}$, perturbative corrections are parametrically small. The problem is not the total absence of an expansion parameter. The problem

is that bare single-stream SPT loop integrals sample internal momenta near and above the nonlinear scale, where the dust equations are incomplete, so part of the coefficient of the long-wavelength expansion is not predicted.

Likewise, “cutoff-dependent in any case” is too strong. For a scale-free spectrum with $n < -1$, the displayed P_{13} UV integral converges as $\Lambda \rightarrow \infty$. It can still receive physically uncontrolled contributions from $q \sim k_{\text{NL}}$, but that is distinct from mathematical divergence.

A better heading is “**Why bare single-stream SPT is not a complete long-wavelength theory.**” State separately:

- the external-momentum expansion can be controlled at small k ;
- internal hard modes are not correctly described by the pressureless fluid;
- the resulting hard contribution is analytic in k and must be matched by EFT operators;
- near k_{NL} the perturbative hierarchy itself also ceases to be useful.

7. Correct the locality statement in the EFT discussion

Location: p. 10, paragraph before Eq. (2.12).

Severity: Must fix.

Small-scale dynamics has finite spatial extent, which supports a spatial derivative expansion. Its response time is not generally short compared with a Hubble time. The EFTofLSS stress tensor is generically nonlocal in time along the fluid trajectory. At fixed perturbative order, this time nonlocality can be represented by a finite set of operators with independent coefficients.

The current sentence first motivates a local-in-time expansion and only later admits time nonlocality in a comment. Reverse this: state nonlocality from the outset, then explain why it does not enlarge the equal-time one-loop power-spectrum basis beyond the listed operators.

8. Fix Appendix B: the displayed stress tensor is schematic and misses a^{-2}

Location: p. 18, Eq. (B.1).

Severity: Must fix.

In comoving coordinates with conformal time, the gravitational stress carries a factor $1/a^2$. The exact kinetic contribution after smoothing is also not simply $\rho u_i^s u_j^s$; it is the difference between the smoothed momentum flux and the flux of the mass-weighted long velocity. A safe schematic form is

$$\tau_{ij}^{\text{kin}} = [\rho u_i u_j]_{\Lambda} - \rho_{\ell} u_{\ell i} u_{\ell j},$$

$$\tau_{ij}^{\text{grav}} = \frac{1}{8\pi G a^2} ([2\partial_i \Phi \partial_j \Phi - \delta_{ij} (\nabla \Phi)^2]_{\Lambda} - \text{extlong} - \text{fieldpiece}).$$

If the appendix intends only a short-short illustration, label it explicitly as schematic and retain the a^{-2} factor.

9. Give the higher-derivative galaxy-bias term a free coefficient

Location: p. 13, Eq. (3.1) and the “Higher derivatives” paragraph.

Severity: Must fix.

Writing $R_*^2 \nabla^2 \delta$ assigns the operator a fixed unit coefficient and a fixed sign in Fourier space. Symmetry fixes the operator, not its coefficient. Use

$$b_{\nabla^2 \delta} R_*^2 \nabla^2 \delta$$

or a dimensionful coefficient $c_{\nabla^2 \delta} \nabla^2 \delta$. Then the contribution is $-b_{\nabla^2 \delta} R_*^2 k^2 P_L$, with an unconstrained sign. Write the composite operators as renormalized operators, for example $[\delta^2]$ and $[\mathcal{G}_2]$, once renormalization has been introduced.

10. Correct the interpretation of the Kaiser degeneracy

Location: p. 14 after Eq. (3.6); p. 17 and solution p. 5 after Exercise 3.3.

Severity: Must fix.

The quadrupole-to-monopole ratio measures $\beta = f/b_1$. The linear full-shape amplitude does not, by itself, determine f and b_1 separately from the primordial amplitude. In the usual parameterization it constrains combinations close to $f\sigma_8$ and $b_1\sigma_8$, together with shape and distance information. A value of f is obtained only within a cosmological model and with an amplitude calibration or prior.

Replace “measuring f itself needs the full shape” and “full shape breaks the $f - b_1$ degeneracy” by a statement in terms of $f\sigma_8$, $b_1\sigma_8$, and model assumptions.

11. Add the survey measurement operator to the end-to-end recipe

Location: pp. 15-17, Secs. 3.4-3.5.

Severity: Must fix relative to the stated goal.

The output of Eq. (3.15) is not yet the object compared to a realistic survey data vector. Add a step of the form

$$P_{i\ell}^{\text{model}} = \sum_{\ell'} \int dk' W_{i\ell,\ell'}(k') P_{\ell'}^{\text{AP}}(k') + P_{i\ell}^{\text{IC}} + P_{i\ell}^{\text{syst}},$$

with the exact form depending on the estimator. Discuss:

- finite k -bin averaging;
- survey-window mixing of k and multipoles;
- the radial/angular integral constraint;
- fiber assignment or collision treatment;
- redshift evolution and effective-redshift/light-cone averaging;
- covariance estimation and finite-mock corrections;
- nuisance templates and priors.

If using a window-deconvolved estimator, explain that one still needs its response, cuts, and residual systematic treatment. This material can be one slide and one boxed paragraph; it is indispensable for an “end-to-end” claim.

12. Add an EFT exercise and fully solve Exercise 3.2

Location: exercises after Lectures 2 and 3; solution set pp. 4-5.

Severity: Major improvement.

Lecture 2 devotes approximately half its content to EFT but none of its exercises tests EFT. Replace or demote the one-dimensional ZA proof and add a cutoff-matching exercise: regulate P_{13} at Λ , derive the running of $c_s^2(\Lambda)$ needed to keep the power spectrum invariant at order $k^2 P_L$, and identify the finite renormalized coefficient.

The current solution to Exercise 3.2 classifies the terms in words but does not reproduce Eq. (C.1). A complete solution should explicitly expand

$$\delta_s^{(2)} = \delta_g^{(2)} - \frac{ik\mu}{\mathcal{H}} u_z^{(2)} - \frac{ik\mu}{\mathcal{H}} [u_z^{(1)} \delta_g^{(1)}] - \frac{1}{2} \left(\frac{k\mu}{\mathcal{H}} \right)^2 [u_z^{(1)} u_z^{(1)}],$$

insert the Fourier velocity relation, symmetrize under $1 \leftrightarrow 2$, and recover every factor of $1/2$, f , μ_i , and k_i in Z_2 .

Detailed audit of the lecture notes

Front matter, audience, and scope

1. **Major improvement - the learning goal overpromises.** “Compute the one-loop EFT prediction” suggests that students can derive or implement the model from the notes. The appendices do not provide

Z_3 , the full bias-loop basis, numerical regularization details, or a complete survey likelihood. Change the goal to “derive the anatomy of, and evaluate with a supplied code, the one-loop prediction.”

2. **Major improvement - the stated prerequisites are insufficient for the planned pace.** Students are assumed to know linear relativistic perturbation theory, but Lecture 1 also expects immediate fluency with kinetic theory, Fourier convolutions, Gaussian random fields, Wick contractions, fluid perturbation theory, and loop power counting. Supply a two-page pre-course primer or a short prerequisite problem set.
3. **Major improvement - distinguish the framework from one implementation.** One-loop EFT full-shape modeling is indeed central to contemporary analyses, but current pipelines use several Eulerian and Lagrangian organizations and different nuisance bases. Rephrase “This is the model” as “This is one standard Eulerian organization of the one-loop EFT framework.”
4. **Moderate improvement - state the scope assumptions in a box on p. 1.** At minimum list: Gaussian adiabatic initial conditions; a single pressureless clustering fluid; scale-independent linear growth and EdS kernels; no primordial non-Gaussianity; no multi-fluid relative operators; plane-parallel RSD; no selection effects; one tracer and one effective redshift; and no survey-window operator until the final step.
5. **Moderate improvement - use $\mathcal{H}$ rather than a calligraphic-looking H that can be confused with physical H .** Keep the explicit identity $\mathcal{H} = aH$ in a notation table.
6. **Minor/editorial - make the Fourier-convention warning operational.** State explicitly whether $P_{\text{here}} = (2\pi)^3 P_{\text{Ref.}[2]}$ or the inverse, rather than only saying they differ by factors of $(2\pi)^3$.

Lecture 1

Section 1.1: relativistic to Newtonian

7. **Moderate improvement - qualify “only scalars matter.”** It is correct for the linear density sector under discussion, but vectors and tensors can source scalar observables at higher order. Add “at the order and for the observables considered here.”
8. **Moderate improvement - define the quoted neutrino 15 percent.** It is not clear whether this means the fractional potential slip, the correction to one potential, or the free-streaming fraction. Say “an order-ten-percent potential slip for the standard neutrino fraction during radiation domination” or give the exact ratio being quoted.
9. **Moderate improvement - call Eq. (1.5) the linear Newtonian system.** The phrase “the Newtonian fluid equations” immediately before displaying linear equations can be confused with the nonlinear Eqs. (1.8)-(1.9).
10. **Major improvement - soften “all gauges agree” on subhorizon scales.** Gauge differences in the relevant density and velocity variables are suppressed by powers of $\mathcal{H}/k$; they do not become literally identical under every definition. State the suppression rather than an absolute equality.
11. **Must fix for internal consistency - the transfer-function paragraph conflicts with the preceding gauge comment.** The statement that larger-scale modes grow as a throughout matter domination is not true for the Newtonian-gauge density while a mode remains superhorizon. Say instead that modes entering after equality avoid radiation-era suppression and, once subhorizon in matter domination, grow approximately as a .
12. **Moderate improvement - label $k_{\text{eq}} \simeq 0.015 h \text{ Mpc}^{-1}$ as a fiducial value.** It depends on $\Omega_m h^2$ and the radiation density.
13. **Moderate improvement - distinguish the transfer function from the full late-time linear spectrum.** A Boltzmann code supplies transfer functions and growth for several species. The lecture later assumes one pressureless field; explain whether P_L means total matter or CDM+baryon power, especially if neutrino masses are varied.

Section 1.2: Vlasov and fluid closure

14. **Moderate improvement - state the cold, collisionless, single-species assumptions before Eq. (1.7).** The exactness claim before shell crossing excludes primordial velocity dispersion, baryonic pressure, and multi-fluid effects.

15. **Moderate improvement - replace “completely wrong after” shell crossing.** The dust equations cease to be closed after shell crossing, but their long-wavelength solutions can remain useful after renormalization. “Not a valid microscopic closure after shell crossing” is more accurate and sets up EFT better.
16. **Moderate improvement - state the nonlinear vorticity result precisely.** With initially potential flow, vorticity remains zero in the single-stream pressureless system before shell crossing. After coarse-graining, an effective vorticity sector can be generated but does not enter the one-loop density power spectrum independently at the order discussed.
17. **Minor/editorial - the symbol f is used for both phase-space density and growth rate.** The joke helps, but a script f_{ps} or F for the phase-space distribution would reduce cognitive load for the target audience.
18. **Moderate improvement - present $f \simeq \Omega_m^{5/9}$ as one fit, not a law.** The commonly used growth-index form is $f \simeq \Omega_m^\gamma$ with γ close to 0.55 in flat LambdaCDM. State the domain and numerical nature of the approximation.

Section 1.3: statistics

19. **Must fix for precision - homogeneity and isotropy are conflated.** Statistical homogeneity yields $\delta_D(\mathbf{k} + \mathbf{k}')$; statistical isotropy makes P a function of $k = |\mathbf{k}|$ only.
20. **Moderate improvement - account for the reality condition.** For a real field, $\delta(-\mathbf{k}) = \delta(\mathbf{k})^*$, so the statement that different wavevectors are uncorrelated needs the exception enforced by $\mathbf{k}' = -\mathbf{k}$.
21. **Moderate improvement - acknowledge primordial non-Gaussianity as an excluded extension.** One sentence is enough: if the initial field is non-Gaussian, connected primordial correlators add terms and induce additional bias operators such as the scale-dependent local-PNG response.
22. **Pedagogical improvement - use one explicit finite-volume mode example.** Students often understand Wick contractions better if the Dirac delta is first replaced by a Kronecker delta in a periodic box.

Section 1.4: Eulerian perturbation theory

23. **Must fix - normalize the velocity divergence and repair Eqs. (1.18)-(1.19).** See highest-priority correction 1.
24. **Moderate improvement - decompose F_2 into growth, tidal, and advection operators rather than assigning the raw μ^2 term entirely to tides.** The trace-free tidal kernel is $\mu^2 - 1/3$; the isotropic and tidal pieces are linear combinations of the constant and μ^2 terms.
25. **Moderate improvement - add the symmetry notation $F_n^{(s)}$.** The recursion produces unsymmetrized kernels, while all displayed loop formulas use symmetrized kernels. Either attach a superscript (s) consistently or state once that F_n henceforth always means the symmetrized kernel.
26. **Moderate improvement - separate two different large-scale statements.** Hard short modes generate a stochastic density amplitude beginning at k^2 , hence a stochastic power beginning at k^4 . A response proportional to a pre-existing long mode produces the deterministic $k^2 P_L$ counterterm. The current “cannot generate power beyond k^4 ” wording invites confusion with the EFT counterterm.
27. **Moderate improvement - “UV modes decouple” needs a qualifier.** They decouple only as powers of k/q in the long-wavelength expansion; their coefficients remain sensitive to nonlinear short-scale physics and require matching.
28. **Must fix - the non-EdS multiplicative factor does not apply to the full kernel.** See highest-priority correction 2.
29. **Moderate improvement - distinguish the EdS approximation for density and velocity.** Replacing $-\mathcal{H}$ by $-\mathcal{H}f$ is part of the usual approximation; it is not the exact time dependence of every higher-order velocity kernel.

Section 1.5: one loop

30. **Moderate improvement - P_{13} does affect shape.** Although it is proportional to $P_L(k)$, the loop coefficient is k dependent. Call it a propagator-like or response term rather than saying it corrects growth but not shape.

31. **Pedagogical improvement - show one complete contraction with all delta functions on one line.** The prose is correct, but a compact displayed derivation would make the origin of the factors 2 and 6 much easier to retain.
32. **Moderate improvement - call the $k = 0$ self-contraction a zero-mode/tadpole subtraction.** Explain that fixing the mean density removes it.
33. **Must fix conceptually - preserve exact IR cancellation while explaining finite BAO displacements.** See highest-priority correction 3.
34. **Moderate improvement - use “equivalence principle” or “extended Galilean invariance” rather than only “Galilean invariance.”** The time-dependent uniform displacement relevant in cosmology is more general than a constant-velocity Galilean boost.
35. **Moderate improvement - the loop-counting statement in powers of Δ^2 is heuristic for LambdaCDM.** It is exact as a scaling statement in a scale-free, local-in-momentum problem. In LambdaCDM the slope varies and UV counterterms form an independent derivative expansion.
36. **Moderate improvement - label the numerical nonlinear scale as convention dependent.** State the cosmology, redshift, whether P_L or nonlinear P is used, and the definition. Figure 3 later uses a different k_σ , which is correctly noted; a small comparison box would help.
37. **Pedagogical improvement - use an IR-safe representation in the numerical companion.** Evaluating P_{22} and P_{13} separately can produce severe cancellation error. Demonstrate a symmetrized integrand that combines the soft regions before numerical integration.

Lecture 1 exercises

38. **Moderate improvement - Exercise 1.1 is too long for the lecture’s core objective.** Keep it as a pre-class or optional derivation. The non-affine conformal-time point is useful, but it consumes solution space without reinforcing loop perturbation theory.
39. **Good and retain - Exercise 1.2 directly supports the lecture.** Add a second check, $F_2(\mathbf{q}, -\mathbf{q}) = 0$, to connect to mass conservation.
40. **Good but revise - Exercise 1.3 is central.** After showing the strict soft cancellation, ask students to explain why a feature with scale k_{osc} still receives finite enhanced displacement corrections.

Lecture 2

Section 2.1: Lagrangian picture and ZA

41. **Moderate improvement - define the reference density associated with $\mathbf{q}$.** Equation (2.3), $1 + \delta = 1/J$, assumes $\mathbf{q}$ labels an asymptotically homogeneous reference configuration. For a finite initial time with nonzero $\delta_i(\mathbf{q})$, the numerator is $1 + \delta_i(\mathbf{q})$.
42. **Must fix - Fig. 2 extends ZA beyond its domain.** See highest-priority correction 4.
43. **Major improvement - “bulk motion essentially exactly” is too strong.** ZA resums linear displacements and captures the leading IR bulk-flow physics nonperturbatively. It is exact in one dimension before shell crossing, not in general three-dimensional dynamics.
44. **Moderate improvement - distinguish a caustic from a virialized object.** The divergence of the ZA density marks failure of the map, not the formation of a stable halo.
45. **Moderate improvement - explain why low-order LPT resums Eulerian terms without implying higher dynamical accuracy at all scales.** The Jacobian creates all powers of the linear displacement, but only a restricted subset of dynamical nonlinearities is included.

Section 2.2: 2LPT

46. **Major improvement - “pure tidal invariant” is inaccurate.** The source

$$\frac{1}{2} \left[(\nabla^2 \phi^{(1)})^2 - \phi_{,ij}^{(1)} \phi_{,ij}^{(1)} \right]$$

is the second principal invariant of the deformation tensor. It contains both an isotropic density-squared part and a trace-free tidal part. The nonzero spherical-patch result in the solution set itself demonstrates this.

- 47. **Must fix - define $f_2 = d \ln |D_2| / d \ln a$.** See highest-priority correction 5.
- 48. **Moderate improvement - state whether $\phi^{(1)}$ and $\phi^{(2)}$ include their time dependence.** Eqs. (2.7)-(2.9) can be read either way. Put the temporal factors in D_1, D_2 and define time-independent spatial potentials, or state explicitly that the displayed potentials are full-time fields.
- 49. **Moderate improvement - qualify the 2LPT initial-condition claim.** 2LPT strongly reduces transients relative to ZA, but the residual depends on starting redshift, force resolution, and the statistic. “Standard” is correct; “percent-level artifacts” should be attached to a stated setup or left qualitative.
- 50. **Pedagogical improvement - make the exact Fourier-map derivation the primary route.** The alternative formula

$$\delta(\mathbf{k}) = \int d^3q e^{-i\mathbf{k}\cdot\mathbf{q}} (e^{-i\mathbf{k}\cdot\Psi} - 1)$$

is cleaner and reduces sign mistakes. Use the Jacobian/transport decomposition afterward for physical interpretation.

Section 2.3: why bare SPT is incomplete

- 51. **Must fix - rewrite the no-expansion-parameter claim.** See highest-priority correction 6.
- 52. **Must fix - remove “cutoff-dependent in any case.”** Distinguish a divergent UV integral from finite but untrustworthy sensitivity to $q \sim k_{\text{NL}}$.
- 53. **Major improvement - “virialized structures should decouple altogether” is too absolute.** Their internal structure affects long modes through multipole- and derivative-suppressed operators. This is decoupling, not zero influence.
- 54. **Moderate improvement - distinguish failure of the microscopic closure from failure of the long-wavelength expansion.** This distinction is the conceptual reason EFT works.

Section 2.4: EFTofLSS

- 55. **Moderate improvement - the mode-count argument needs assumptions.** $N \sim V k_{\text{max}}^3$ and errors $\propto N^{-1/2}$ assume an approximately Gaussian covariance, homogeneous selection, and negligible shot-noise domination. Present it as scaling intuition.
- 56. **Moderate improvement - do not promise that EFT automatically prevents bias.** Power counting gives an expected scaling of omitted terms with unknown order-one coefficients. Scale cuts still require validation on simulations or controlled mock challenges.
- 57. **Moderate improvement - a smoothing kernel need not be a spherical top-hat.** Define $[X]_\Lambda = W_\Lambda * X$ and state that physical predictions are independent of the arbitrary filter after matching to the retained order.
- 58. **Moderate improvement - clarify the hierarchy $k \ll \Lambda$ and the role of k_{NL} .** The external modes must be well below the matching scale; the long fields retained below Λ must be perturbative. The final prediction cannot depend on the arbitrary separation scale.
- 59. **Must fix - correct the time-locality motivation.** See highest-priority correction 7.
- 60. **Moderate improvement - distinguish the bare coefficient from the renormalized parameter.** The counterterm does not equal the UV part of P_{13} ; its running cancels cutoff dependence and its finite part is measured.
- 61. **Major improvement - Eq. (2.14) is invariant only up to omitted operators and perturbative order.** For sufficiently blue spectra, P_{22} produces a k^4 UV sensitivity requiring a stochastic/higher-derivative term. Write $\partial_\Lambda P = 0 + O(k^4, \text{higher loops})$ at the retained order.
- 62. **Moderate improvement - “the coefficient is of order P_L ” is bookkeeping, not a literal observable scaling.** Explain that the insertion is assigned one-loop order; the fitted coefficient is a property of short-scale

dynamics and carries dimensions.

63. **Good but sharpen - the matter stochastic power begins at k^4 .** Explain the chain: mass and momentum conservation force the long-wavelength stochastic density to start at k^2 , hence its auto-power starts at k^4 .
64. **Major improvement - “the theory predicts its own error bar” overstates EFT power counting.** It predicts the *parametric scaling* of the error. The coefficient and correlations across k are not known without a model, prior, or simulation calibration.
65. **Moderate improvement - Figure 3 is an illustration, not validation.** It fits one coefficient to halofit using a no-wiggle input. Put “toy illustration” in the caption and avoid quantitative claims of demonstrated percent accuracy from this plot.
66. **Pedagogical improvement - add a cutoff-running figure.** Show $P_{\text{SPT}}(k; \Lambda)$ for several Λ and then the collapse of the curves after fitting/running $c_s^2(\Lambda)$. That figure would teach renormalization more directly than a single halofit comparison.

Lecture 2 exercises

67. **Good - Exercise 2.1 checks the sign convention.** Add the definition $f_2 = d \ln |D_2| / d \ln a$ immediately afterward.
68. **Good - Exercise 2.2 is central.** Ask students to identify the trace-free tidal part explicitly, not only the constant, dipole, and μ^2 terms.
69. **Moderate improvement - Exercise 2.3 needs its domain stated.** It concerns plane-parallel, collisionless sheets with a homogeneous-background subtraction and appropriate periodic or asymptotic boundary conditions, not arbitrary one-dimensional point-particle gravity.
70. **Major improvement - add an EFT exercise.** The absence of any EFT calculation is the largest mismatch between learning objectives and assessment.

Lecture 3

Section 3.1: galaxy bias

71. **Moderate improvement - distinguish formation nonlocality in space and time.** R_* controls spatial derivatives; time nonlocality is along the past fluid trajectory and does not have to be short compared with H^{-1} .
72. **Moderate improvement - qualify the velocity-bias statement.** The equivalence principle forbids a leading, scale-independent velocity bias for freely falling conserved tracers. Higher-derivative velocity bias, tracer non-conservation, and line-of-sight selection effects are allowed.
73. **Must fix - add a free higher-derivative coefficient.** See highest-priority correction 9.
74. **Major improvement - write renormalized composite operators explicitly.** After the renormalization paragraph, Eq. (3.1) should use brackets such as $[\delta^2]$ and $[\mathcal{G}_2]$ or clearly state that all operators are renormalized by definition.
75. **Moderate improvement - explain that the omitted cubic operators are degenerate only for the one-loop two-point function under the stated assumptions.** They matter independently for higher-point functions and other observables.
76. **Moderate improvement - $b_1 = \lim_{k \rightarrow 0} P_{gm}/P_{mm}$ assumes Gaussian initial conditions and no scale-dependent large-scale response.** State the scope; local primordial non-Gaussianity is the standard counterexample.
77. **Must fix in notation - the stochastic expansion should be additive.** Write

$$P_{\epsilon\epsilon}(k) = P_{\epsilon}^{\{0\}} + P_{\epsilon}^{\{2\}} k^2 + O(k^4),$$

rather than $P_{\text{shot}}[1 + O(k^2 R_*^2)]$, because the k^2 coefficient is an independent parameter and need not scale with the constant.

- 78. **Moderate improvement - do not say different patches are literally uncorrelated.** Short-scale stochasticity has a finite correlation length, which implies analyticity at low k .
- 79. **Moderate improvement - label coevolution relations as model priors.** They follow from a restricted conserved-tracer evolution model and are not universal halo identities.
- 80. **Moderate improvement - add selection operators to the scope box.** Real target selection can depend on line-of-sight quantities and introduces operators not contained in an isotropic real-space bias expansion.

Section 3.2: redshift-space distortions

- 81. **Moderate improvement - state the validity of the plane-parallel approximation.** Wide-angle, curved-sky, relativistic, and evolution terms are omitted. These are usually small on the quasi-linear scales considered but matter on the largest scales or for very wide surveys.
- 82. **Must fix - interpret the Kaiser information as $f\sigma_8$ and $b_1\sigma_8$.** See highest-priority correction 10.
- 83. **Moderate improvement - specify that Z_3 is symmetrized in Eq. (3.9).** The appendix says this later; the main formula should not rely on the reader finding that warning.
- 84. **Moderate improvement - the counterterm basis is convention dependent.** Eq. (3.10) follows a CLASS-PT-style parameterization. State explicitly that the coefficients already absorb combinations of matter EFT, higher-derivative bias, and redshift-space short-scale responses, and give the mapping to the per-multipole c_0, c_2, c_4 basis if that is what the likelihood uses.
- 85. **Moderate improvement - call the $k^4\mu^4$ term an enhanced formally higher-order proxy.** It is not part of the minimal one-loop order in naive power counting; it is retained when line-of-sight velocity dispersion makes it numerically important.
- 86. **Moderate improvement - do not identify all FoG physics with a convergent derivative expansion.** At sufficiently small $k\mu\sigma_v$, FoG can be expanded into EFT operators. At larger values, phenomenological damping or another reorganization may be more efficient, but then it must be consistently matched and validated.
- 87. **Moderate improvement - specify the degeneracy in Eq. (3.11).** The $a_2\mu^2k^2$ stochastic term is nearly degenerate with the enhanced $\tilde{c}k^4\mu^4Z_1^2P_L$ contribution over a limited range, not generically with every redshift-space counterterm.
- 88. **Minor/editorial - list dimensions for a_0 and a_2 .** They have dimensions of length to the fifth power in the displayed convention.
- 89. **Moderate improvement - mention that one-loop $P(k, \mu)$ contains higher even multipoles.** The loop polynomial through μ^8 generates $\ell = 6, 8$. Even if the data vector uses only $\ell = 0, 2, 4$, higher multipoles can mix into them through AP and the survey window.

Section 3.3: IR resummation

- 90. **Must fix conceptually - preserve exact $q \rightarrow 0$ cancellation.** See highest-priority correction 3.
- 91. **Moderate improvement - explain the role of k_S .** It is an arbitrary long/short separation scale. Residual k_S dependence at finite order is a higher-order error estimate and a useful implementation check.
- 92. **Moderate improvement - distinguish r_d , r_{BAO} , and k_{osc}^{-1} .** They are closely related but not identical definitions. The current text acknowledges a small difference; a one-line definition table would prevent students treating them as exactly interchangeable.
- 93. **Major improvement - include counterterms in the resummed formula consistently.** The main formula omits them and only the end-to-end recipe later says production codes damp the wiggly part of the counterterm. Put the consistent prescription in the main section or explicitly label Eq. (3.14) as the tree-plus-loop core before EFT and stochastic completion.
- 94. **Moderate improvement - identify the approximation in the wiggly-loop factorization.** The formula neglects or approximates terms with multiple wiggly insertions and depends on the wiggle/no-wiggle algorithm. These errors are higher order but should be named.

Section 3.4: multipoles, AP, and parameters

95. **Moderate improvement - use the comoving angular diameter distance D_M for the transverse mapping.** Ratios of D_A at fixed redshift are numerically equivalent because the factors of $1+z$ cancel, but D_M matches the comoving coordinates used throughout.
96. **Major improvement - include the AP volume Jacobian in the displayed equation.** It is part of the model, not an optional parenthetical detail.
97. **Major improvement - add bin averaging and the survey window after Eq. (3.15).** See highest-priority correction 11.
98. **Moderate improvement - Table 1 values are highly convention and sample dependent.** Label them as illustrative prior scales for one BOSS-like convention, not typical physical values. State whether the quoted P_{shot} is the full constant or the residual after Poisson subtraction.
99. **Moderate improvement - include all nuisance parameters that appear in the displayed model.** Table 1 omits a_0 and a_2 from Eq. (3.11). Either add them and their units or state that the baseline drops them.
100. **Moderate improvement - separate physical bias parameters from purely phenomenological/analysis parameters.** A compact grouping into bias, derivative/RSD counterterms, and stochastic terms would be clearer.
101. **Major improvement - scale cuts are not set only by crossing a theoretical-error curve.** They are selected using power counting together with mock/simulation validation, estimator systematics, prior robustness, and parameter-bias tests.

Section 3.5: loops and inference

102. **Moderate improvement - FFTLog is an implementation, not the conceptual completion of the theory.** Modern full-shape pipelines also use configuration-space Hankel methods, Lagrangian resummations, emulators, and other factorizations. Present FFTLog as one efficient route.
103. **Moderate improvement - “the transform converges for any tilt” is numerically misleading.** A finite discrete transform exists for any chosen bias exponent, but aliasing, endpoint periodicity, dynamic range, and the convergence strip of the analytic loop integral strongly constrain a stable implementation.
104. **Major improvement - add the convergence domain and analytic continuation of Eq. (D.3).** Students should know that the Gamma-function formula is initially derived in a convergence strip and then analytically continued, with care near poles.
105. **Major improvement - mention padding, windowing, tilt choice, and IR/UV subtractions.** Without these, Appendix D is insufficient to reproduce reliable numbers despite the claim that the appendices are enough to do so.
106. **Moderate improvement - a Boltzmann code does not uniquely output a wiggle/no-wiggle split.** It outputs P_L and growth quantities; the split is a separate filtering or template procedure with algorithmic uncertainty.
107. **Must fix relative to the stated end-to-end goal - insert the survey measurement operator between AP/multipoles and likelihood.** See highest-priority correction 11.
108. **Moderate improvement - distinguish marginalization from ignoring parameters.** Marginalization integrates or sums over nuisance directions with their priors; simply dropping or fixing them is not equivalent.
109. **Moderate improvement - note covariance assumptions.** A Gaussian residual likelihood with a fixed covariance is an approximation. If covariance is estimated from a finite mock ensemble, its uncertainty modifies the likelihood or inverse-covariance treatment.
110. **Moderate improvement - do not imply that counterterms safely absorb arbitrary missing loops.** They absorb contributions with the shapes allowed by the retained operator basis. Missing non-analytic or linearly independent higher-order shapes cannot be assumed away.
111. **Major improvement - qualify the BOSS result.** The cited analysis is independent of CMB *data*, but its baseline uses a BBN prior on ω_b and fixes or constrains the primordial tilt and neutrino sector. Say this explicitly.
112. **Major improvement - update the contemporary context.** Add DESI DR1 full-shape results and

the comparison of velocileptors, PyBird, and FOLPS-like one-loop EFT models. This makes clear that the lecture teaches a common framework rather than a unique code basis.

113. **Moderate improvement - “a chain needs 10^5 evaluations” is implementation dependent.** Modern samplers, analytic nuisance marginalization, emulators, and gradient-based methods can change this by orders of magnitude. The key point is repeated fast evaluation, not a fixed count.

Appendices

114. **Appendix A - use the normalized velocity variable introduced in the main text.** Otherwise G_n appears without a fully explicit connection to θ .
115. **Appendix A - state that the recursion shown is for unsymmetrized ordered kernels.** The final kernels are averages, not sums, over all permutations; this is already said but should be visible before the equations or in their label.
116. **Appendix B - correct the stress tensor.** See highest-priority correction 8.
117. **Appendix C - avoid relying only on typeface to distinguish G_2 the kernel from $\mathcal{G}_2$ the bias operator.** Use G_2 for the velocity kernel and $\mathcal{G}_2$ for the Galileon operator everywhere, including plain-text descriptions and code.
118. **Appendix C - a reference to Z_3 is not enough for the stated “compute” goal.** Either provide the explicit symmetrized $Z_3(\mathbf{q}, -\mathbf{q}, \mathbf{k})$ configuration needed by P_{13} , provide a code/notebook, or weaken the learning objective.
119. **Appendix D - the redshift-space loops are not obtained by a naive scalar monomial expansion alone.** The text does mention tensor master integrals, but it should emphasize that line-of-sight contractions and bias structures require a nontrivial matrix basis.
120. **Appendix D - add a numerical-validation checklist.** Compare against direct integration for selected cosmologies, test independence of FFTLog tilt and grid range, test low- k cancellations, and verify the $k \rightarrow 0$ limits.

Detailed audit of the solution set

The solutions are concise and generally correct. Their main weakness is not algebraic error but uneven completeness: the early exercises are fully worked, while the most complicated redshift-space exercise is only sketched.

1. **Exercise 1.1 - correct.** The non-affine conformal-time term and Hubble-drag check are useful. Add an explicit perturbative-order statement explaining that Φu is second order in the weak-field/velocity expansion. The final comment should say that the geodesic reduction itself does not require $k \gg \mathcal{H}$, while identifying Φ through a Poisson equation does.
2. **Exercise 1.2 - correct.** Use $F_2^{(s)}$ and $G_2^{(s)}$ consistently. Add the zero-total-momentum check $F_2(\mathbf{q}, -\mathbf{q}) = 0$.
3. **Exercise 1.3 - algebra correct, final interpretation needs revision.** The factors from the two soft regions and the angle-averaged F_3 are handled correctly. Replace “it does not cancel for sharp features” with the finite- q/k_{osc} explanation given above.
4. **Exercise 2.1 - correct.** The spherical-patch discussion is pedagogically valuable and directly shows why the phrase “pure tidal invariant” in the notes is misleading. Add the corrected definition of f_2 .
5. **Exercise 2.2 - correct but convention-sensitive.** State explicitly whether the spatial potentials are time independent and where D_+ and D_2 have been absorbed. The exact Fourier-map derivation should be promoted from an optional fallback to the main derivation, followed by the four-term decomposition as interpretation.
6. **Exercise 2.3 - correct under unstated boundary conditions.** Specify plane-parallel sheets, homogeneous-background subtraction, and a reference point or periodic zero mode that fixes the integration constant in $\int \delta dx = -\Psi$.
7. **Exercise 3.1 - correct.** Retain the sign check. It is one of the strongest short solutions in the set.
8. **Exercise 3.2 - incomplete.** It identifies the four physical sources of Z_2 but does not perform the requested derivation. Supply the explicit convolution algebra, symmetrization, and coefficient checks.

9. **Exercise 3.3 - algebra correct, interpretation needs revision.** Replace the claim that full shape breaks the $f - b_1$ degeneracy with the correct statement about $f\sigma_8$, $b_1\sigma_8$, power-spectrum shape, and priors.
10. **Major assessment gap.** No solution addresses EFT renormalization, stochastic power counting, renormalized bias operators, IR-resummation bookkeeping, AP plus Jacobian, or window convolution. At least three of these should become required problems.
11. **Pedagogical improvement - add common-error callouts.** Useful examples: forgetting the mirror soft region in P_{22} ; confusing G_2 and $\mathcal{G}_2$; taking $\ln D_2$ with $D_2 < 0$; omitting the AP Jacobian; and treating $1/\bar{n}$ as a fixed stochastic prediction.
12. **Pedagogical improvement - add numerical checks.** A five-line numerical test for each analytic result would substantially improve retention and expose convention errors.

Recommended three-lecture design

The present notes should be assigned as a pre-read. The live lectures should teach the conceptual spine and one or two derivations per lecture, not every displayed formula.

Lecture 1: From the fluid to one loop

- **0-10 min:** relativistic-to-Newtonian bridge; what is assumed from the Boltzmann code.
- **10-22 min:** Vlasov moments, single-stream closure, linear growth.
- **22-32 min:** Gaussian fields and one Wick-contraction example.
- **32-55 min:** normalized velocity divergence, SPT expansion, derivation and interpretation of F_2 .
- **55-70 min:** P_{22} and P_{13} , diagrams, pairing factors.
- **70-80 min:** IR safety, UV sensitivity, and the precise question EFT must answer.
- **80-90 min:** conceptual check, questions, and summary.

Move the geodesic derivation and full kernel recursion to pre-class/appendix material.

Lecture 2: Lagrangian dynamics and EFT completion

- **0-18 min:** displacement map, mass conservation, ZA, first shell crossing.
- **18-33 min:** 2LPT and the contact with F_2 through the exact Fourier map.
- **33-43 min:** why bare dust loops are UV incomplete.
- **43-63 min:** coarse-graining, mass-weighted velocity, effective stress, spatial derivative expansion, time non-locality.
- **63-75 min:** one-loop counterterm and cutoff independence.
- **75-82 min:** stochastic k^4 term and power counting/theory error.
- **82-90 min:** cutoff-running mini-exercise and summary.

Make the one-dimensional exactness proof optional.

Lecture 3: Galaxies, RSD, BAO, and the data vector

- **0-22 min:** symmetry basis for bias, renormalized operators, stochasticity, higher derivatives.
- **22-42 min:** redshift-space map, Kaiser limit, anatomy of Z_2 and Z_3 .
- **42-55 min:** one-loop EFT spectrum and nuisance sectors.
- **55-67 min:** IR resummation and anisotropic BAO damping.
- **67-78 min:** AP mapping, multipoles, binning, higher-multipole mixing, survey window.
- **78-83 min:** end-to-end likelihood pipeline and what the data constrain.
- **83-90 min:** summary, limitations, and current pipelines.

Move the FFTLog derivation to the companion note and a notebook. Keep only the power-law-factorization idea in the live lecture.

Recommended exercise set

Lecture 1

1. **Required:** derive $F_2^{(s)}$ and check the soft and zero-total-momentum limits.
2. **Required:** derive the P_{22} - P_{13} leading soft cancellation and rewrite the numerical integrand in an IR-safe form.
3. **Optional/pre-class:** conformal-time geodesic derivation.
4. **Numerical:** plot P_{22} , P_{13} , and their sum for a toy power law and for LambdaCDM.

Lecture 2

1. **Required:** derive F_2 from the Lagrangian Fourier map.
2. **Required:** regulate the one-loop UV contribution and derive the running of $c_s^2(\Lambda)$ at order $k^2 P_L$.
3. **Required conceptual:** use mass and momentum conservation to derive stochastic $P_{\text{stoch}} \propto k^4$ for matter.
4. **Optional:** prove 1D ZA exactness before shell crossing.

Lecture 3

1. **Required:** derive Kaiser multipoles and identify the measured combinations $b_1 \sigma_8$ and $f \sigma_8$.
2. **Required:** derive the full Z_2 kernel from the redshift-space map.
3. **Required:** renormalize δ^2 at low k and explain how its cutoff-dependent piece shifts b_1 .
4. **Required numerical:** implement the wiggle/no-wiggle damping and vary k_S .
5. **Required practical:** apply AP plus a supplied window matrix to theory multipoles and compare with the unwindowed result.

Suggested replacement passages

Replacement for the SPT expansion

Define the dimensionless velocity divergence $\Theta \equiv -\theta/(\mathcal{H}f)$. In the EdS-kernel approximation,

$$\delta = \sum_{n \geq 1} D_+^n \delta_n, \quad \Theta = \sum_{n \geq 1} D_+^n \Theta_n,$$

where δ_n and Θ_n are convolutions of the initial field with the time-independent symmetrized kernels F_n and G_n . Outside EdS the exact kernels have weak time dependence; retaining EdS kernels with the exact D_+ and f is an accurate but not exact approximation.

Replacement for the IR paragraph

The poles of F_n and G_n for a soft leg represent a coherent displacement by a long mode. The strict $q \rightarrow 0$ contribution cancels in equal-time correlators, as required by the equivalence principle. A feature such as BAO introduces an additional scale k_{osc} : modes can be soft relative to the hard mode while having q/k_{osc} of order unity. The resulting finite IR-enhanced displacement terms smear the feature and must be resummed.

Replacement for the SPT-failure paragraph

At external wavenumbers $k \ll k_{\text{NL}}$, perturbation theory is still an expansion in small long-wavelength fluctuations. Bare SPT is nevertheless incomplete because its loops integrate over internal momenta where the pressureless single-stream equations are invalid. The hard contribution is analytic in the external momentum and is not predicted by the dust theory. EFTofLSS supplies all symmetry-allowed long-wavelength operators whose coefficients are fixed by matching, rendering predictions independent of the arbitrary hard-mode cutoff to the retained order.

Replacement for the EFT-locality paragraph

Integrating out short modes produces a response that is local in space only as a derivative expansion and is generally nonlocal in time along the long fluid trajectory. At a fixed perturbative order, the time dependence can be represented by a finite operator basis with free coefficients. The scalar part begins with operators proportional to δ , θ , and higher spatial derivatives.

Replacement for the bias expansion

$$\delta_g = b_1[\delta] + \frac{b_2}{2}[\delta^2] + b_{\mathcal{G}_2}[\mathcal{G}_2] + b_{\Gamma_3}[\Gamma_3] + b_{\nabla^2\delta}R_*^2\nabla^2\delta + \epsilon + \dots$$

Brackets denote renormalized composite operators. The coefficients are responses of the tracer abundance to long-wavelength backgrounds. The stochastic field has an analytic low- k spectrum, $P_{\epsilon\epsilon} = P_{\epsilon}^{\{0\}} + P_{\epsilon}^{\{2\}}k^2 + \dots$.

Replacement for the Kaiser-information sentence

The ratio P_2/P_0 depends only on $\beta = f/b_1$. Including the amplitude gives the combinations $f\sigma_8$ and $b_1\sigma_8$. Inferring f or b_1 separately requires a cosmological model and an amplitude calibration or prior; broadband shape and higher-order statistics add information but do not remove this logical dependence.

Replacement for the end-to-end recipe

1. Compute the linear CDM+baryon or total-matter spectrum appropriate to the chosen neutrino treatment, together with D_+ and f .
2. Construct and validate the wiggle/no-wiggle split.
3. Evaluate tree, loop, bias, counterterm, and stochastic basis spectra.
4. Apply IR resummation consistently to all wiggly pieces.
5. Apply the AP mapping including its Jacobian and project all multipoles needed for subsequent mixing.
6. Average over the data bins and apply the survey/estimator response: window matrix or deconvolution response, integral constraints, and observational-systematic templates.
7. Evaluate the likelihood with the correct covariance treatment and priors; validate scale cuts with simulations or blinded mock challenges.
8. Sample or optimize the posterior, marginalizing rather than fixing nuisance directions unless explicitly justified.

Visual and editorial review

1. **Overall design is strong.** The typography is clean, equations are legible, hyperlinks and comments are visually distinct, and the figures are integrated rather than decorative.
2. **The pages are too dense for live teaching.** Pages 10, 13-17, and 19 contain long uninterrupted text/equation blocks. For lecture use, provide a slide or board companion with one conceptual statement and one equation per visual unit.
3. **Figure 1 is effective.** Add arrows for external momentum and a small legend distinguishing an external line from an integrated loop line.
4. **Figure 2 must be physically relabeled.** Use “formal eigenvalue sequence” after the first caustic and do not label the final object “halo” without a warning.
5. **Figure 3 should be labeled a toy comparison.** Halofit and a fitted one-parameter no-wiggle curve are not a precision validation of EFTofLSS.
6. **Figure 4 is useful.** Add the no-wiggle baseline explicitly and state whether Σ is one-dimensional or the convention entering $e^{-k^2\Sigma^2}$.
7. **Figure 5 is useful but the legend is slightly misleading.** “Tree level (Kaiser, IR-resummed)” is better called “IR-resummed leading order.” State why kP_ℓ rather than P_ℓ is plotted.

8. **Table 1 is too compressed.** Split it into three groups - bias, counterterms, stochasticity - and put convention-dependent prior scales in a separate column or footnote. Include all terms actually used in the displayed model.
9. **Add a one-page notation map.** Include $\delta^{(n)}$, δ_n , F_n , G_n , Θ , $\mathcal{G}_2$, Γ_3 , Φ_g , Φ_v , μ , μ_i , D_A/D_M , and every counterterm basis.
10. **Reduce absolute language.** Replace “completely wrong,” “essentially exactly,” “predicts its own error bar,” and “everything else comes out correct” with statements tied to perturbative order and assumptions.

References and contemporary context to add

The current bibliography is solid through the CLASS-PT/BOSS era but should be updated and slightly rebalanced.

1. Add a direct reference for EFT in redshift space and the origin of the counterterm basis, rather than relying mainly on CLASS-PT for Eq. (3.10).
2. Add a standard modern LPT reference in addition to a dissertation.
3. Add a reference on bias renormalization/operator bases and, if space permits, selection effects.
4. Add DESI 2024 V, *Full-Shape Galaxy Clustering from Galaxies and Quasars* (arXiv:2411.12021).
5. Add the DESI comparison of velocileptors, PyBird, and FOLPS-like EFT models (arXiv:2404.07272).
6. Add a recent independent DESI DR1 EFT power-spectrum-plus-bispectrum analysis (arXiv:2507.13433) as an example of the expanded modern data vector and systematic treatment.
7. Mention simulation- or HOD-informed nuisance priors as an active extension rather than presenting broad top-hat/Gaussian nuisance priors as the only current practice.
8. Improve reference [8] with full bibliographic information or a stable URL/version.
9. Clarify the status and availability of the companion note [21]. If it is required to reproduce results, distribute it and its code together with these notes.

Final prioritized revision checklist

Before the lectures

- Repair Eqs. (1.18)-(1.19) using a normalized velocity divergence.
- Correct the non-EdS F_2 statement.
- Rewrite the strict-IR versus BAO discussion in both notes and solutions.
- Relabel Fig. 2 and remove physical post-shell-crossing claims.
- Define $f_2 = d \ln |D_2| / d \ln a$.
- Rewrite Sec. 2.3 around UV incompleteness, not absence of any expansion parameter.
- Correct time nonlocality and Appendix B’s a^{-2} factor/exact stress definition.
- Add the coefficient $b_{\nabla^2 \delta}$ and use renormalized bias operators.
- Correct all claims about measuring f to statements about $f\sigma_8$ and assumptions.
- Add survey binning/window/systematics to the final recipe.
- State the priors behind the quoted BOSS cosmology result.
- Fully solve Exercise 3.2 and add an EFT renormalization exercise.

Strongly recommended

- Add a scope/assumptions box and a notation table.
- Reallocate each lecture to 80 minutes of content plus 10 minutes of questions/recap.
- Move FFTLog derivation and geodesic details to reading/notebooks.
- Add a cutoff-running figure and a survey-window example.
- Update the current-analysis discussion to DESI and multiple one-loop pipelines.
- Provide a tested numerical notebook with IR-safe loops, BAO damping, AP, and window convolution.

Optional polish

- Split Table 1 by nuisance sector.

- Add common-error boxes to solutions.
- Replace absolute wording throughout.
- Add explicit finite-volume interpretations of Dirac deltas and Wick contractions.

Bottom line

The draft is scientifically ambitious, well organized, and close to being an excellent compact set of specialist notes. Its core formulas are mostly sound. The present weaknesses are concentrated in a manageable set of conceptual overstatements, two genuine notation/formula problems, an unrealistic time budget, and the missing measurement operator between theoretical multipoles and survey data. Fixing those points would raise the notes from a strong internal draft to a credible mini-course that experts could recommend without caveat.